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Bidirectional learning in upbound and downbound microzones of the cerebellum

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Abstract | Over the past several decades, theories about cerebellar learning have evolved. A relatively simple view that highlighted the contribution of one major form of heterosynaptic plasticity to cerebellar motor learning has given way to a plethora of perspectives that suggest that many different forms of synaptic and non-synaptic plasticity, acting at various sites, can control multiple types of learning behaviour. However, there still seem to be contradictions between the various hypotheses with regard to the mechanisms underlying cerebellar learning. The challenge is therefore to reconcile these different views and unite them into a single overall concept. Here I review our current understanding of the changes in the activity of cerebellar Purkinje cells in different 'microzones' during various forms of learning. I describe an emerging model that indicates that the activity of each microzone is bound to either increase or decrease during the initial stages of learning, depending on the directional and temporal demands of its downstream circuitry and the behaviour involved.

Control system
A system that regulates the behaviour of an animal using control loops.

Training
The process of instructions, rehearsal and feedback that leads to learning.

Cerebellar learning
The induction of a lasting alteration in cerebellar behavior or the potential thereof.

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https://doi.org/10.1038/s41583-020-00392-x

Why is cerebellar learning important? Orchestrating our sensorimotor behaviours and cognitive functions requires us to be capable of controlling our neuronal activity at high temporal resolution[1]. For example, preparing for the return of an approaching tennis ball or baseball while deciding on the precise direction, speed and curve one wants to achieve requires a control system that can efficiently integrate all relevant internal and external parameters and come up with a solution in a time frame on the order of tens of milliseconds. The required levels of online integration and temporal precision are so high that such a control system can function well only if it is endowed with vast learning capabilities that allow us to react in a manner that is based on extensive training in the past. While the mammalian cerebral cortex is well connected to all other parts of the brain and provides critical command signals[2,3], the properties of its output neurons — the pyramidal cells — are not well suited to control the rest of the brain at high temporal resolution. Pyramidal cells fire at a relatively low baseline frequency, and individual pyramidal cells often project over long distances to many different downstream circuitries[4], hampering ultraprecise spatial and temporal control. The cerebellar cortex, on the other hand, appears to be an excellent candidate to precisely control the activity of its outputs over time in a learning-dependent manner[5–10]. Its sole output neuron, the Purkinje cell, fires at a high spontaneous firing frequency (between 50 and 120 Hz), allowing precise bidirectional changes in firing frequency to be made that facilitate cerebellar learning[11,12]. Moreover, both the dendrites and the axons of Purkinje cells are endowed with structural and biophysical features that facilitate specific spatial and temporal control[13–15]. This allows cerebellar Purkinje cells to control motor or non-motor functions while exploiting the timing-sensitive feedback mechanisms integrated within the olivocerebellar system (FIG. 1).

What is a microzone? In the vertebrate cerebellar cortex, Purkinje cells are embedded into well-segregated microzones, sagittal strips of cells that are each involved in the control of a specific autonomic, motor or cognitive function[16–21] (FIG. 1). The specificity of the actions of microzones is supported by their targeted projections: each microzone projects to a particular set of cerebellar and/or vestibular nuclei, together establishing a microcomplex[21–23]. Even though the cerebellar and vestibular nuclei comprise a bewildering palette of different projection neurons and local interneurons[24–27], the dendrites of which can extend even into neighbouring regions[28], the baseline activity patterns of the main projection neurons in these nuclei appear to adhere to the organization of the microcomplexes[29]. Moreover, the cerebellar and vestibular nuclei of individual microcomplexes in turn inhibit the specific subnucleus of the inferior olive from which the corresponding Purkinje cells receive their climbing fibre input[30–32]. As a result,

the olivocerebellar system as a whole comprises many parallel micromodules that are formed by topographically organized three-element loops, comprising specific sets of Purkinje cells, neurons in the cerebellar and/or vestibular nuclei and inferior olive neurons[31,33–35] (FIG. 1b).

The function, and thereby identity, of each microzone and micromodule can be determined by the specific responses of its Purkinje cells to the climbing fibre activity that occurs following a particular form of sensorimotor stimulation or internal activation[13,17,18,36,37]. Specific olivocerebellar microzones are, for example, activated by particular forms of peripheral stimulation, including selective stimulation of the hindlimb or forelimb nerves, a puff onto the eyelid or the presentation of a visual stimulus to the eye[17,23,38–42]. Likewise, other micromodules have been suggested to be activated by internal activation, such as that occurring during reward estimation, planning or decision-making[43–47].

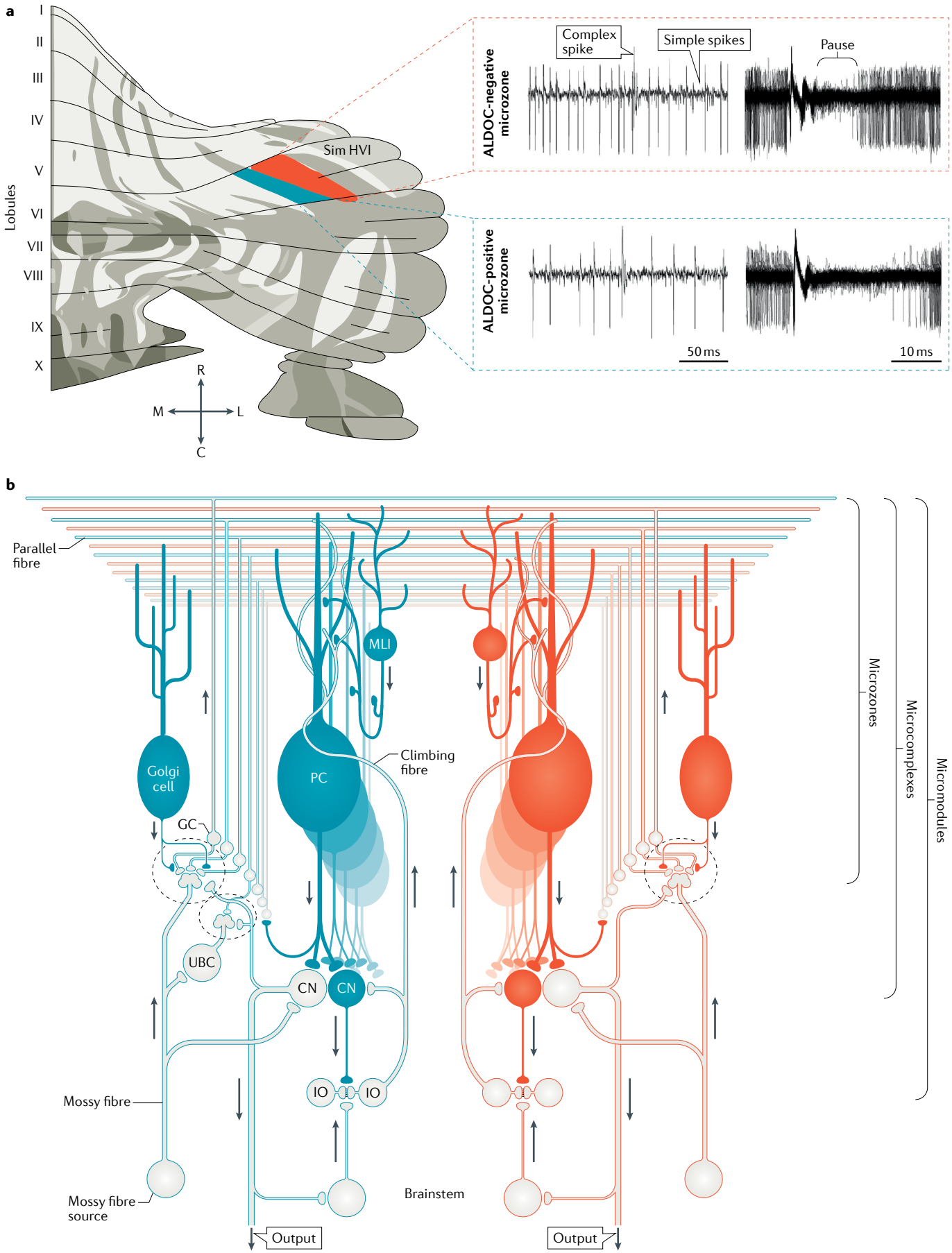
The spikes that are triggered by climbing fibres in the Purkinje cells are the so-called complex spikes[48,49]; these all-or-none responses usually occur at a low baseline firing frequency of 1 Hz, and this does not rise much higher than 7–11 Hz following stimulation[13]. They therefore stand out against the simple spikes of the Purkinje cells, which can fire at frequencies much greater than 100 Hz (FIG. 1a). Although this simple spike activity is sustained by intrinsic mechanisms, it can be modulated through activation of brainstem neurons giving rise to the mossy fibre–parallel fibre pathway as well as by different types of interneurons located in the molecular layer of the cerebellar cortex[50]. Whereas the axons of molecular layer interneurons are — like the climbing fibres and the dendritic trees of Purkinje cells — oriented in the sagittal plane, parallel fibres run perpendicularly, traversing many different microzones. As a result, the cerebellar cortex forms a modular matrix, enabling specific changes in the synaptic strength of the inputs from the parallel fibres and interneurons onto Purkinje cells to occur under the guidance of enhanced or reduced climbing fibre activity[51–53].

What is the challenge? Over the past half century the prevailing view has been that cerebellar motor learning in vertebrates depends largely on long-term depression (LTD) at the parallel fibre to Purkinje cell synapse[52,54]. This synaptic change can be induced when parallel fibres and climbing fibres are activated simultaneously or within a relatively narrow time frame[55–58]. However, evidence has been accumulating that this synapse can also be subject to long-term potentiation (LTP) and that many other forms of synaptic and non-synaptic plasticity also occur in the cerebellar cortex as well as downstream in the cerebellar and vestibular nuclei[53,59–68]. It has also been shown that the Purkinje cells of different microzones often show opposite changes in their simple spike activity during the expression of learned behaviours[69]. Even the changes in their climbing fibre input during learning do not always appear consistent, with both increases and decreases observed during learning as well as altered responsiveness to multiple sensory modalities and prominent transitions in activity during the training process itself[37,70–76]. Moreover, recent studies have shown that the contribution of climbing fibre activity to both motor and non-motor function extends beyond the transmission of error signals (as classically assumed) to the relay of reward and the signalling of the expectation thereof[43–47,77–79]. Finally, whereas the canonical cytoarchitecture of olivocerebellar modules was well defined by the classical anatomy of the previous century[20,80,81] (FIG. 1a), the zonal differences in the intrinsic electrophysiological and molecular properties of its neurons have been revealed only over the past decade[11]. So, how can we explain these enigmatic findings and resolve the potential contradictions? An important challenge in the field is to integrate the newer findings with the older information to generate one overall unifying model of the mechanisms underlying cerebellar learning.

Microzone directionality. As this Review discusses, the first key to an overall understanding of cerebellar learning may be provided by growing evidence that the neurons in each cerebellar microzone change their activity in one particular direction that is optimal for a specific type of behaviour: that is, by either increasing or decreasing their firing rate during the initial stages of training. For example, mammals appear to use different microzones for learning a compensatory eye movement around the vertical axis in a clockwise direction as compared with an anticlockwise direction[40,82,83], they may engage different microzones over time during the expression of conditioned eyeblink closures and the subsequent eyelid opening[72,84–88], and they may use partly different microzones for planning to initiate or abort a specific motor programme following object discrimination[43,44,78,89]. In each of these studies, the simple spike activity of the Purkinje cells of each microzone appears to change in a single direction (either upbound or downbound) during the acquisition of a particular type of behaviour[69].

The second key to understanding how the learning cerebellum may operate is the notion that the upbound or downbound nature of the microzone involved in a particular behaviour can probably be predicted by the net polarity of neurotransmissions in the downstream circuitry that mediates it. More specifically, given the inhibitory nature of Purkinje cells[90], they can be expected to participate in an upbound or downbound microzone if the net polarity of the downstream circuitry is inhibitory or excitatory, respectively. In this way, the target neuron at the end of the downstream circuit that mediates the initial primary effect will, in general, be automatically excited. One advantage of an arrangement with such bidirectional features is that cerebellar control of the specific behaviour involved is always possible, regardless of the net polarity of synaptic transmissions in the downstream network involved. Another advantage may be that microzones that exhibit opposing activity changes can operate in a push–pull fashion during learning, allowing fine-tuning of complementary functions simultaneously or in a particular sequence at high temporal resolution.

In this Review, I describe the evidence for the existence of upbound and downbound learning microzones, consider the extent to which their character can be predicted by polarity of neurotransmission in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Lobules
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ALDOC-positive microzone
50 ms
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b
Parallel fibre
MLI
Golgi cell
PC
Climbing fibre
GC
UBC
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CN
IO
IO
Mossy fibre
Mossy fibre source
Brainstem
Output
Output
Microzones
Microcomplexes
Micromodules

Fig. 1 | **Organization of the cerebellar system. a** | The lobules of the cerebellar cortex (I–X) are organized in a mediolateral (M–L) direction into different sagittal microzones, many of which can be distinguished by the distribution of particular proteins[94,98] (left panel). In the schematic, the microzones in the mouse cerebellum are visualized on the basis of the expression levels of aldolase C (ALDOC; also known as zebrin II). Two neighbouring microzones within lobulus simplex in the cerebellar hemisphere (Sim HVI) are indicated in blue (ALDOC positive) and red (ALDOC negative). A functional microzone of Purkinje cells can be identified by their responses to their climbing fibre input, which can be activated by a specific form of sensorimotor or internal stimulation. The climbing fibre triggers a complex spike (up deflection) in the Purkinje cell (panels on the right). The infrequent complex spikes stand in marked contrast to the more numerous simple spikes (down deflections), which are generated by intrinsic activity of the Purkinje cells and by their parallel fibre input. The activity profile of a single-unit Purkinje cell can be identified by the occurrence of a pause in simple spike activity directly after the complex spike[13]; the panels on the extreme right show multiple superimposed traces of a single Purkinje cell at higher temporal resolution. Upper and lower panels show examples of baseline activity of Purkinje cells in an ALDOC-negative microzone and an ALDOC-positive microzone, respectively. **b** | The cytoarchitectures of neighbouring microzones, such as those indicated in part **a**, are virtually identical, except that unipolar brush cells (UBCs) reside predominantly in the ALDOC-positive microzones. The intrinsic networks of neighbouring microzones are largely segregated and almost exclusively connected through beams of parallel fibres, which originate from granule cells (GCs)[146]. Microzones are embedded in microcomplexes, which are in turn embedded in micromodules[16–21]. A microzone of inhibitory Purkinje cells (PCs) in the cerebellar cortex is topographically connected to a particular set of cerebellar nucleus neurons (CNs) and/or vestibular nucleus neurons, together forming a microcomplex, while a microcomplex is reciprocally connected with a particular subnucleus of neurons in the inferior olive (IO), together forming a micromodule. The CNs and/or vestibular nucleus neurons provide an inhibitory feedback to electrotonically coupled olivary neurons, which in turn excite the corresponding PCs and molecular layer interneurons (MLIs) through their climbing fibres, closing the structure of the micromodule[11]. Whereas all climbing fibres are derived from the IO, mossy fibres originate from a large variety of sources in the brainstem. The mossy fibres project to glomeruli (dashed ellipses), where they make contact with dendrites of the GCs, Golgi cells or UBCs located in the input stage of the cerebellar cortex. The GCs receive recurrent feedback within the same sagittal microzone from PCs and CNs[23,67,190]; likewise, the PCs and MLIs receive recurrent feedback within the same sagittal microzone from other PCs[191]. Cells and terminals depicted with open and closed structures are excitatory and inhibitory, respectively. Arrows indicate the flow of action potentials. C, caudal; R, rostral. Part **a** adapted from REF.[11], CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/). Part **b** reprinted from REF.[13], Springer Nature Limited.

Microzones
Small, sagittally organized networks of cerebellar Purkinje cells and interneurons dedicated to the control of a specific aspect of autonomic, motor or cognitive function.

Microcomplex
A cerebellar cortical microzone of Purkinje cells and an associated small group of cerebellar nucleus cells that are dedicated to the control of a specific aspect of autonomic, motor or cognitive function.

Micromodules
Microcomplexes of Purkinje cells and cerebellar nucleus cells connected with a small cluster of neurons in the inferior olive that together control a specific aspect of autonomic, motor or cognitive function.

their downstream circuitry and suggest how different microzones may work together. The emerging learning rules will be illustrated by providing various examples, ranging from simple forms of sensorimotor learning to higher-order forms of planning.

Purkinje cell activity in microzones

As noted already, whereas the individual identity of each Purkinje cell microzone is determined by the cells' specific responses to climbing fibre activity (that is, the complex spike responses), microzones as a whole have been suggested to be segregated into two main types on the basis of their predominant changes in simple spike activity during the initial stages of learning[69]. Purkinje cells in upbound microzones are proposed to have a relatively low spontaneous simple spike firing frequency, which increases when downstream events have to be accelerated (FIG. 2a). Purkinje cells in downbound microzones, on the other hand, are suggested to have a relatively high spontaneous simple spike firing frequency, which decreases when downstream events have to be accelerated (FIG. 2b). It is hypothesized that both the upbound and the downbound changes in simple spike activity can be supported by opposite changes in the activity of the molecular layer interneurons[72,91,92] (FIG. 2).

The evidence for segregated microzones of Purkinje cells that show differences in the baseline firing frequency of their simple spike activity is robust[11,93]. This phenomenon has been shown in both mice and rats. It holds when excitatory and/or inhibitory afferents innervating the Purkinje cells are genetically or pharmacologically blocked, it occurs in vitro as well as in vivo (in both anaesthetized and awake preparations) and it is also reflected in the patterns of simple spike activity that occur directly after a complex spike[11,94]. The difference in baseline simple spike firing in different microzones is mainly founded in the differential intrinsic expression patterns of many types of different proteins, several of which are related to the control of membrane potential and cellular excitability[94–98]. The two markers that are most widely used for identifying the different zones in mammals are the enzymes aldolase C (ALDOC; also known as zebrin II) and PLCB4; these proteins themselves contribute only indirectly to the differences in Purkinje cell activity[53,69,97] but provide the most conspicuous immunocytochemical labelling for most of the upbound and downbound microzones, respectively (FIG. 3). In addition, differences in the responses to climbing fibre activity in upbound and downbound micromodules may contribute to the differences in baseline simple spike activity; climbing fibre terminals in upbound microzones are likely to release more glutamate per action potential, to have larger pools of release-ready vesicles and to exhibit enhanced multivesicular release and larger synaptic glutamate transients[99]. This will result in longer-duration complex spikes and thereby temporarily prolong the simple spike interval[99].

Whereas systematic differences in the baseline firing frequency of simple spikes among different microzones have already been described for the entire cerebellar cortex in rodents[11], concomitant unidirectional changes in activity during learning have so far only been reported for a limited number of microcomplexes controlling specific types of behaviours[69]. However, the picture emerging from the literature, which is steadily expanding, appears consistent. In general, the simple spike activity of Purkinje cells that initially fire at a low baseline firing frequency increases during training (hence, I suggest that these should be termed 'upbound microzones'). This increase occurs specifically in the periods of the paradigm during which complex spike activity is relatively low[73,100,101]. The increased simple spike firing results, at least in part, from a combination of reduced complex spike activity and enhanced activity in the mossy fibre–parallel fibre pathway, which induces LTP at the parallel fibre to Purkinje cell synapse and LTD at the parallel fibre to molecular layer interneuron synapse[53,56] (FIG. 2a). The resulting increase in simple spike activity, which exerts extra inhibition from the Purkinje cells onto the neurons in the cerebellar and/or vestibular nuclei[90], is in line with the required changes in the downstream target regions[73,101]. To date, the experimental evidence reviewed here[22,27,29,40,73,101] indicates that the training-enhanced acceleration of the activity of a particular muscle group is achieved through activation of an upbound microzone

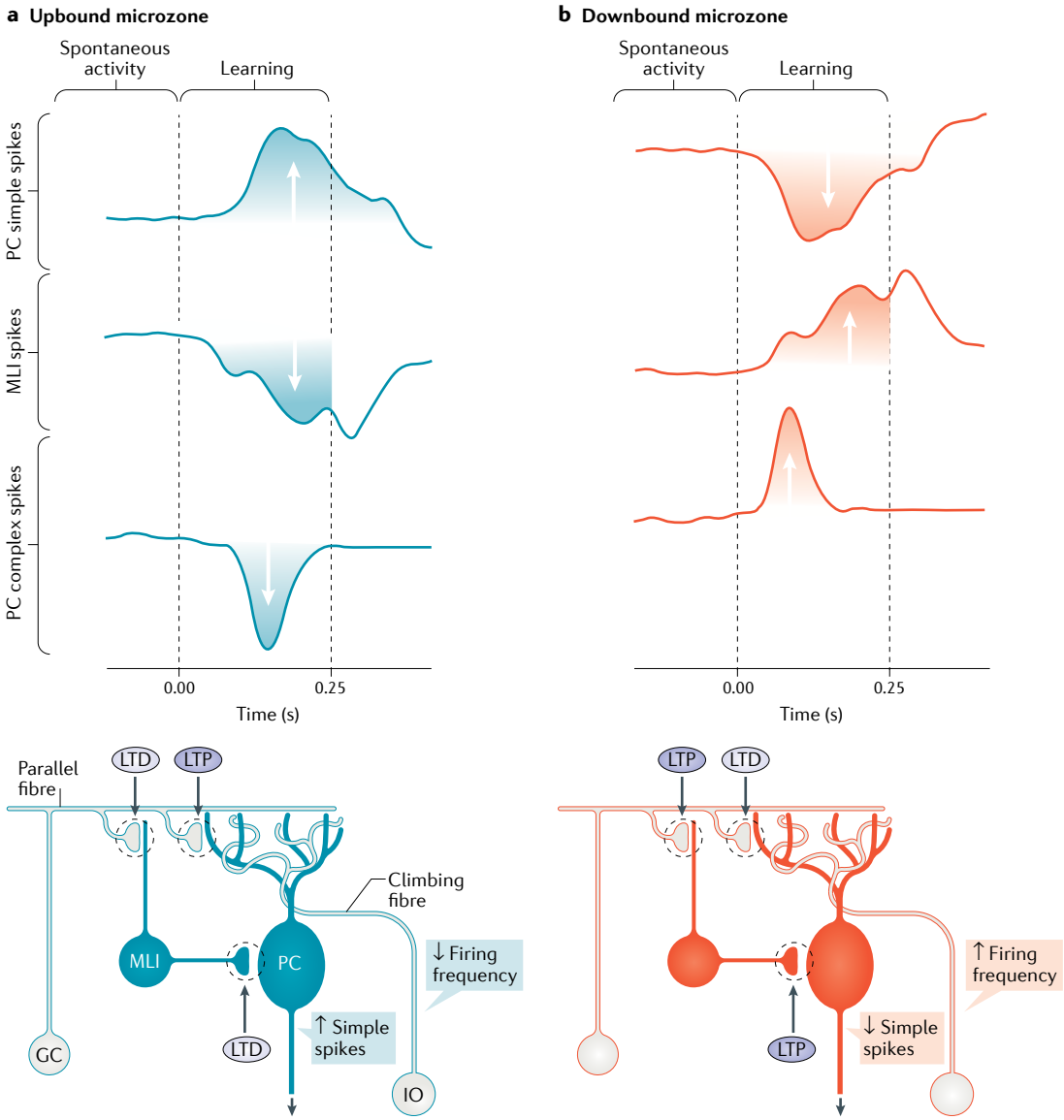


Fig. 2 | **Upbound and downbound microzone activity.** The schematics illustrate the relationships between the dominant unidirectional changes in spike activity and the induction of major types of plasticity during learning for upbound and downbound cerebellar microzones. **a** | Purkinje cells (PCs) in upbound microzones have a relatively low simple spike firing frequency during spontaneous activity, and this activity increases during the initial stages of learning. This is partly due to decreased activity in the molecular layer interneurons (MLIs) and climbing fibres, the latter of which leads to a reduction in PC complex spikes. The images in the top panel represent drawings of 'typical' responses rather than actual data. At the cellular level, these changes in spike activities can be explained by the different forms of synaptic plasticity that are synergistically controlled by climbing fibre activity (bottom panel)[53,64,91]. When climbing fibre activity decreases, the PC will be subject to induction of long-term potentiation (LTP) at the parallel fibre to PC synapse, as well as to long-term depression (LTD) at the parallel fibre to MLI synapse and LTD at the MLI to PC synapse. **b** | PCs in downbound microzones show a relatively high simple spike firing frequency during spontaneous activity, and this activity decreases during learning. The downbound decrease in simple spike activity is driven by increases in activity of the MLIs and complex spike activity of the PCs. These changes in spike activities are congruent with the known forms of synaptic plasticity induced by increased climbing fibre activity; these include enhanced LTD at the parallel fibre to PC synapse, as well as enhanced LTP at the parallel fibre to MLI synapse and the MLI to PC synapse[53,91]. Granule cells (GCs) and neurons in the inferior olive (IO) give rise to the parallel fibres and climbing fibres, respectively.

if there is an odd number of inhibitory neuronal connections in the downstream circuitry of the Purkinje cells of that particular microzone (FIG. 4a).

By contrast, in downbound microzones the simple spike activity of Purkinje cells starts at a high baseline firing frequency and decreases during training epochs in which complex spike activity is relatively high and the temporal interval between the different training stimuli is within the proper range[43,71,72,102–106]. The simple spike suppression arises due to a combination of an increased number of well-timed complex spikes and enhanced activity in the mossy fibre–parallel fibre

Complex spikes
The low-frequency all-or-none spikes of a Purkinje cell that are evoked by activity in its climbing fibre input.

Simple spikes
The high-frequency spikes of a Purkinje cell that are intrinsically generated or evoked by activity in its parallel fibre input.

Long-term depression
(LTD). A process involving postsynaptic changes that renders the synapse less sensitive to an input.

pathway, which synergistically induce LTD at the parallel fibre to Purkinje cell synapse, LTP at the parallel fibre to molecular layer interneuron synapse and potentiation of the molecular layer interneuron to Purkinje cell synapse[53,56,107] (FIG. 2b). The resulting change in the simple spike activity of the Purkinje cells in the downbound microzones also appears well in line with the required changes in the downstream target regions[108,109]. The simple spike suppression leads to reduced inhibition of the neurons in the cerebellar nuclei as well as a training-enhanced acceleration of a particular muscle group, because there are no or an even number of inhibitory neuronal connections in the downstream circuitry of the Purkinje cells of that particular downbound microzone (FIG. 4b).

Thus, the evidence suggests that whether a Purkinje cell participates in an upbound or a downbound microzone determines its propensity for certain forms of plasticity and aligns with the net polarity of the synaptic connections in its downstream circuitry. Together, the different upbound and downbound microzones may allow new combinations of downstream effects to be achieved (such as the simultaneous or consecutive acceleration of contractions of two complementary muscle

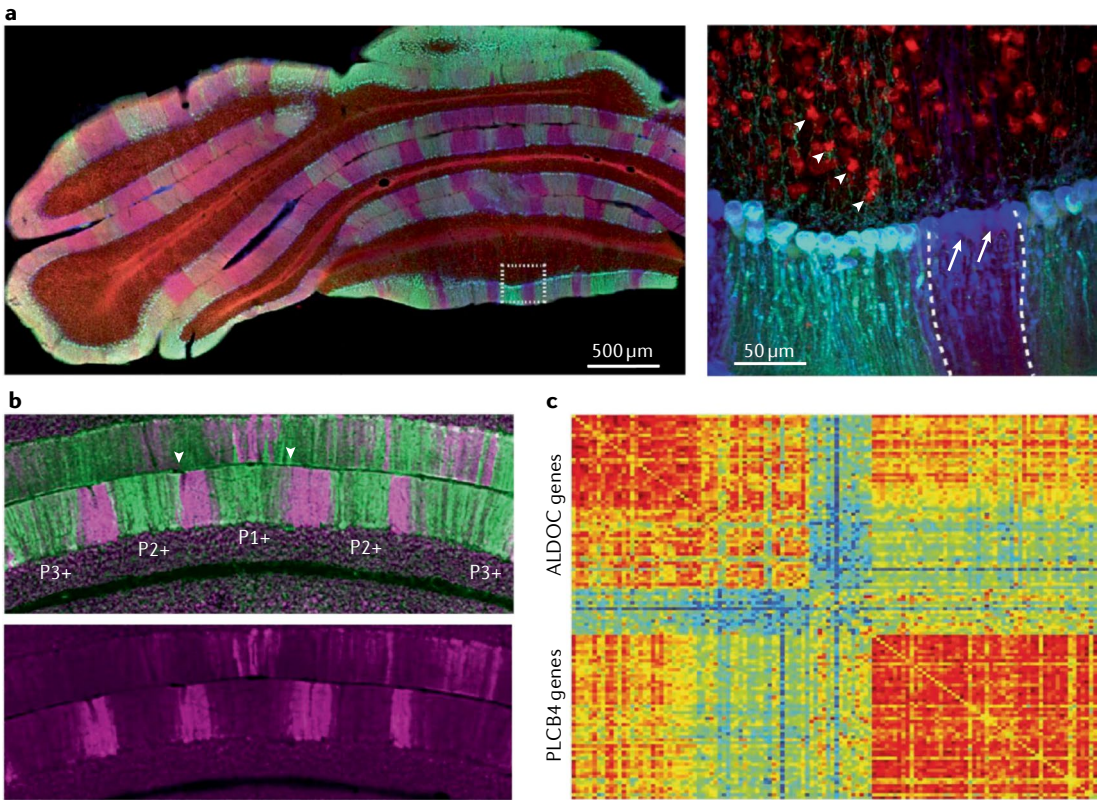


Fig. 3 | **Molecular characteristics of microzones.** Purkinje cells (PCs) in upbound and downbound microzones express different combinations of proteins. **a** | Most upbound microzones prominently express aldolase C (ALDOC; also referred to as zebrin II)[69,73,101,179], whereas many downbound microzones prominently express transient receptor potential cation channel type C3 (TRPC3), controlling the intrinsic excitability of their PCs[98]. In this immunofluorescence image of a coronal section of the mouse cerebellar cortex, TRPC3, ALDOC and calbindin (which labels all PCs) are highlighted with red, green and blue staining, respectively. The higher-magnification image in the right panel corresponds to the box in the left panel. In PCs (white arrows), the expression of TRPC3 largely complements that of ALDOC. By contrast, the expression of TRPC3 in unipolar brush cells (white arrowheads) adheres largely, but not exclusively, to the upbound microzone. **b** | The expression of 1-phosphatidylinositol 4,5-bisphosphate phosphodiesterase-β4 (PLCB4; in magenta in both panels), which is the activator of TRPC3, complements that of ALDOC (green in upper panel) as can be observed here in lobule VIII of the posterior cerebellum of an adult mouse[95]. ALDOC-positive zones are marked as P1+, P2+ and P3+ using the standard zebrin nomenclature[69]. Examples of well-defined microzone boundaries are marked with arrowheads in the upper panel. **c** | Slide-seq analysis of PCs has revealed that the expression patterns of 669 genes are spatially distributed in a non-random manner and that 126 of them appear to be correlated or anticorrelated with the expression patterns of ALDOC or PLCB4 (REF.[97]). The heatmap shown reveals that the 126 Purkinje cell-expressed genes associated with ALDOC or PLCB4 expression can be readily separated into two main clusters (shown in the top half and the bottom half, respectively). The colour coding of the heatmap represents the statistical magnitude of the intersection of correlating genes, with red and blue being the highest and lowest, respectively. Each column or row represents the statistical values for a particular gene. The genes within these datasets include those encoding various potassium channel subunits (such as *Kcnab1*, *Kcnc3*, *Kcng4* and *Kcnma1*), which are likely to boost the intrinsic excitability and firing frequency of PCs in the PLCB4-positive, downbound microzones[11,162,164,166]. In addition, many of the genes may be involved in controlling plasticity in upbound and/or downbound microzones[62,69,94,96,165]. Part **a** adapted from REF.[98], CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/). Part **b** adapted with permission from REF.[95], SFN. Part **c** reprinted with permission from REF.[97], AAAS.

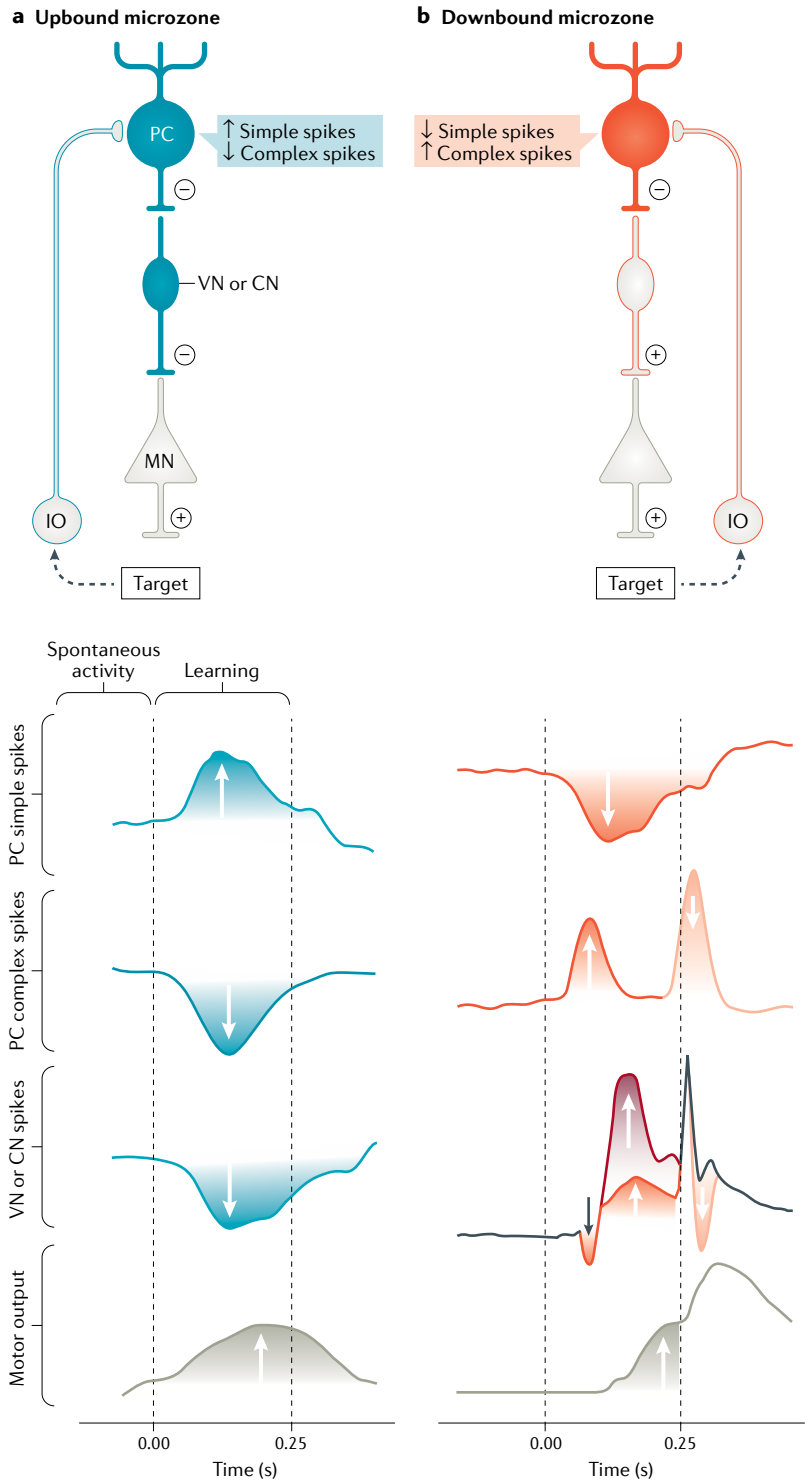


Fig. 4 | **Net polarity of downstream circuitry for upbound and downbound microzones.** The top panels depict hypothesized typical structures of the networks downstream of upbound (left in blue, panel **a**) and downbound (right in red, panel **b**) microzones. The images in the bottom panels represent drawings of 'typical' responses rather than actual data. **a** | The dominant changes in the simple spike activity of upbound microzones during learning appear to be in line with the changes in the activity of downstream target regions that are required for the relevant change in behaviour (such as an increase in the amplitude or acceleration of a movement that is primarily controlled by that microzone)[73,101]. If there are an odd number of inhibitory neuronal connections in the circuitry downstream of the Purkinje cells (PCs), comprising at least one vestibular nucleus neuron (VN) or cerebellar nucleus neuron (CN), next to an effector neuron or a motor neuron (MN), then training-enhanced activation of a particular agonistic function or motor action is predominantly achieved through increases in simple spike activity. The main instruction signal for learning in this direction is a reduction in climbing fibre signals (leading to a reduction in complex spikes)[53,64,91]. The changes in simple spike and complex spike activity will exert their effects on the VNs and/or CNs and thereby on the motor output largely through mechanisms of rate coding. **b** | In downbound microzones, the dominant changes in simple spike activity during learning are also in line with the changes in the downstream target regions that are required for learning in the agonistic direction[69,72]. However, here the polarities of the connections and signals are opposite to those in upbound microzones: if there are no or an even number of inhibitory neuronal connections in the circuitry downstream of PCs, then training-enhanced activation of a particular agonistic muscle group is at least partly achieved through decreases and increases in simple spike and complex spike activity, respectively. Thus, here the feedback mechanisms mediated by the climbing fibre system include a reinforcing form of signalling (akin to reward signalling)[88]. At the level of the cerebellar nucleus, the changes in complex spike and simple spike activity will induce effects not only via rate coding but also via temporal coding, including rebound firing. There are two factors that contribute to rebound activity in the CNs. First, the enhanced climbing fibre-mediated complex spike activity induces a short reduction in CN activity, which in turn triggers a subsequent increase in the activity of the same neurons (effect in red). Second, the simple spike suppression that follows on the increased complex spike activity at the PC level further enhances the increase in CN activity (superimposed effect in dark red). These events occur during the learning epochs in which the simple spike suppression occurs. Outside these learning epochs, other dynamics can occur, which are dependent on the reflexes involved; these could, for example partly, reflect unconditioned reflexes[69,72]. For clarity, the number of connections downstream of the PCs has been kept to a minimum of two. In the real brain this number can be increased to many more for particular upbound and downbound microzones; however, the model predicts that the learning rules hold regardless of these expansions. IO, inferior olive. Panel **b** adapted from REF.[88], CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/).

groups), generating a system that is capable of learning through parallel training mechanisms and simultaneously facilitating temporal lead–lag compensations of different types of motor and non-motor functions at high resolution.

Long-term potentiation (LTP). A process involving postsynaptic changes that renders the synapse more sensitive to an input.

Learning via upbound microzones

There are at least two well-described forms of mammalian cerebellar motor learning that follow the rules of upbound microzones explained earlier: adaptation of the vestibulo-ocular reflex (VOR) and adaptation of reflexive whisker protraction (FIG. 5).

Adaptation of the VOR, which serves to maintain stability of images on the retina during head

movements, is controlled by upbound microzones in the floccular complex of the vestibulocerebellum[40,68,110–113]. This form of learning occurs when the image on the retina during compensatory eye movements is not stable despite consistent vestibular input[113–116], as may occur if one is ageing and starting to wear magnifying lenses[117]. For an increase in the gain (that is, the amplitude) of the horizontal VOR, eye movements around the vertical axis are enlarged by the activation of the abducens motor neurons and the lateral rectus muscle, due to an increase in the simple spike activity of Purkinje cells within the ipsilateral floccular complex[73,92,110,114]. In line with this, the floccular Purkinje cells that control adaptation of the VOR fire at a relatively low baseline firing frequency of about 50–75 Hz during spontaneous activity, but increase their simple spike frequency substantially when the head moves in the contralateral direction, driving the eyes to move towards the ipsilateral side[73,110,114–116]. During adaptation, when the gain of the ipsiversive VOR has to be further increased with respect to that of head rotation, the simple spike activity of the floccular Purkinje cells increases in an approximately linear fashion over time, at least during the initial stages of learning[73,110,117,118] (FIG. 5a). This increase in Purkinje cell activity appears to be sufficient to induce the required behavioural increment, as optogenetic stimulation of the floccular Purkinje cells can drive the eye movements and the amplitude of these eye movements covaries with the intensity of optogenetic stimulation[73].

The notion that the simple spike increases have an eminent role in this gain-increase adaptation of the VOR is also supported by mouse mutant studies, which showed that genetically blocking LTP at the parallel fibre to Purkinje cell synapse induces more prominent behavioural phenotypes than blockage of LTD at the same synapse[64,65,68,92]. Moreover, this difference in severity of phenotypes aligns well with the fact that the effects of the simple spike increases on VOR adaptation are most prominent during the stimulus episode when complex spike firing frequency is low, which renders the Purkinje cells permissive for induction of LTP at their parallel fibre inputs[64,65,68].

The learning-dependent changes in simple spike activity observed during VOR adaptation also align with the downstream circuitry (FIG. 4a); the floccular Purkinje cell target neurons in the vestibular nuclei inhibit the motor neurons in the abducens nucleus that move the eyes into the ipsilateral direction[73,119]. By contrast, the changes in simple spike activity do not align with mossy fibre activity in the circuitry upstream of the floccular Purkinje cells: when the head moves to the contralateral side, the activity in the ipsilateral horizontal semicircular canal as well as that of the neurons in the vestibular nuclei that project to the flocculus decreases[120]. This configuration implies that the phase of the vestibular input signals to the flocculus with respect to the timing of head rotation is reversed during the transformation from sensory coding into simple spike encoding. In other words, by the time the vestibular signals reach the Purkinje cells of the cerebellar cortex, where they are merged with signals from other parts of the oculomotor system, the polarity of the signalling tends to be the opposite to that generated in the vestibular apparatus. Such reversal might be facilitated by inhibition of granule cells provided by GABAergic neurons in the floccular white matter[121], by the actions of one of the main types of unipolar brush cells that has been shown to mediate a phase conversion of excitatory input from the mossy fibres[122] and/or by the phase-shaping impact of the climbing fibres[91].

Adaptation of whisker protraction, which serves to modify whisker reflexes following external perturbation, is controlled largely by upbound microzones embedded in crus 1 and crus 2 of lobule VII of the cerebellar cortex[101,123]. Purkinje cells in these microzones fire at a relatively low baseline firing frequency of about 50–75 Hz, and their simple spike frequency increases substantially when the whiskers protract in response to air-puff stimulation[124,125]. Accordingly, when the amplitude of this protraction is increased, as occurs after a short training period during which the frequency of air-puff stimulation onto the whiskers is increased[101], the simple spike activity further increases in a linear fashion. Consistent with other upbound microzones, such as those responsible for VOR adaptation[73,110], the increases in simple spike activity during learning are most prominent in periods and cells when and where the complex spike responses are relatively weak[89,101]. In addition, here too, optogenetic stimulation of the relevant Purkinje cells is sufficient to further drive the reflexive whisker movements, and mouse mutants that are unable to induce LTP at the parallel fibre to Purkinje cell synapse lack adaptation of whisker protraction[101]. Moreover, the learning-dependent changes in simple spike activity align with the downstream circuitry (FIG. 5b). The target neurons in the cerebellar nuclei most likely modulate the trigeminofacial feedback loop in the brainstem that controls touch-induced whisker protraction via the intermediate reticular formation[126]. This region receives a direct excitatory input from the cerebellar nuclei[127] and inhibits the whisker motor neurons in the facial nucleus[128–130], in line with the prediction that upbound microzones serve to control downstream circuitries with an odd number of inhibitory neuronal connections in their trajectory (FIG. 4a).

It has been shown that a single upbound microzone can contribute even to the control of two different yet related behaviours, such as whisking and respiration, provided that they have the same net polarity in their downstream circuitry[131]. To what extent the mossy fibre network that is activated by sensory whisker manipulation upstream of the cerebellar cortex is out of phase with the Purkinje cell simple spike modulation during adaptation of whisker protraction remains to be elucidated[101]; however, activity in the trigeminal ganglion cells that receive whisker input often increases during air-puff stimulation, and this depends on the direction of stimulation that is used to induce the protraction[132,133] (FIG. 5b).

Learning via downbound microzones

The forms of cerebellar motor learning that can be used to illustrate the learning rules of downbound microzones include Pavlovian eyeblink conditioning and learning how to plan goal-directed movements (FIG. 6).

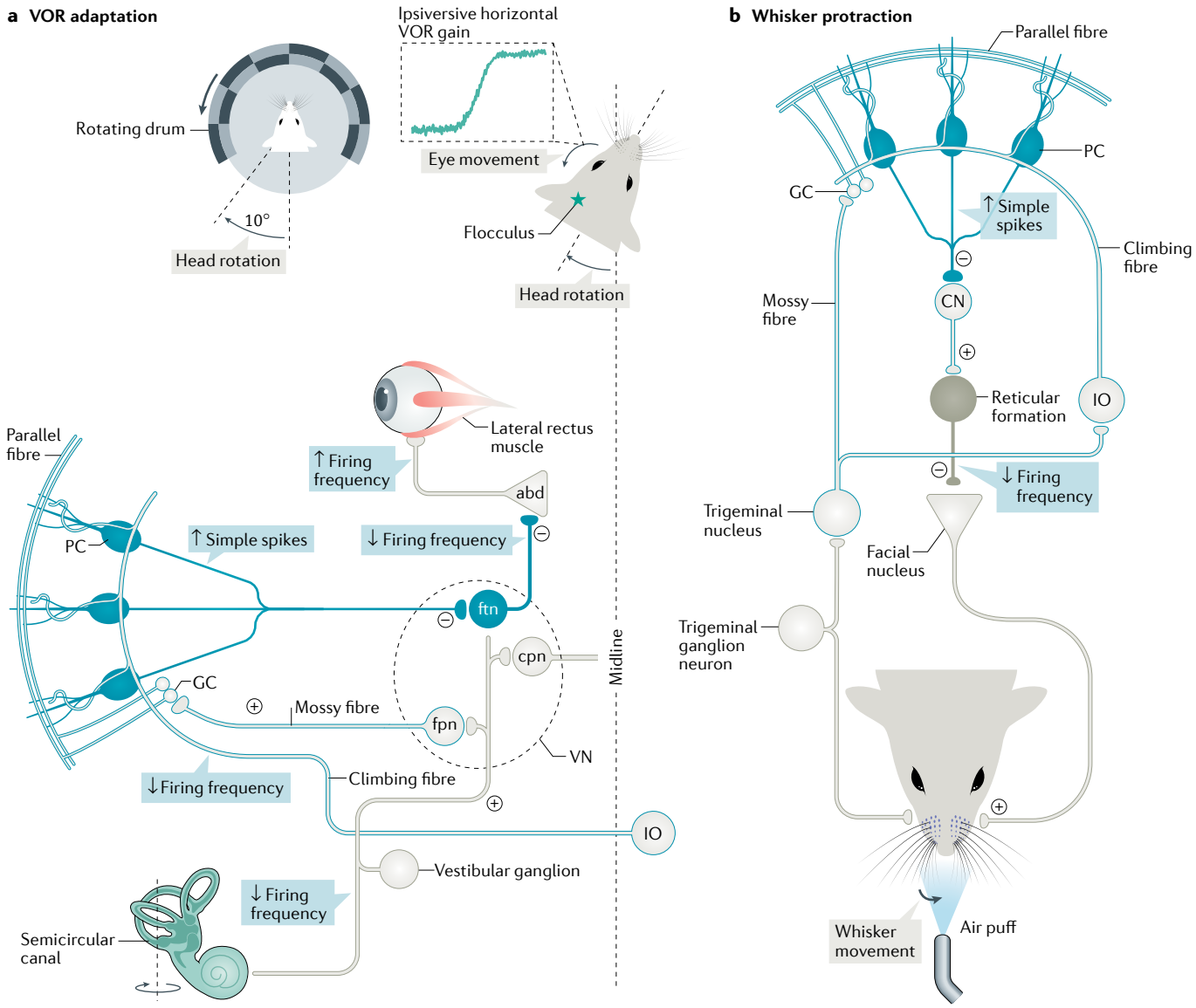
Net polarity of neurotransmissions
The ultimate excitatory or inhibitory outcome of a set of synapses that connect different neurons in series.

Excitability
The propensity of a neuron to generate, beyond a certain threshold, an output signal (the action potential) from a given input signal.

Floccular complex
The combination of the flocculus and the paraflocculus.

Vestibulocerebellum
The region of the cerebellum that receives prominent inputs from the vestibular apparatus and controls behaviours that depend on the balance of the body in space.

Unipolar brush cells
A class of glutamatergic interneurons found in the granular layer of the cerebellar cortex.



Pavlovian eyeblink conditioning, which allows us to generate new associative sensorimotor memories following the systematic pairing of a neutral conditioned stimulus, such as a tone or light, with an unconditioned stimulus, such as an air puff to the eye (FIG. 6a), is predominantly controlled by downbound microzones in the deep fissure of lobule VI in the cerebellar hemisphere (also referred to as 'lobulus simplex' or 'Sim HVI')[38,39,72,85,134]; these microzones are probably integrated in the C3 and D0 modules of the olivocerebellar system[69]. Purkinje cells in this region fire initially at a comparatively high baseline simple spike firing frequency of about 80–110 Hz, which remains relatively constant when spontaneous eyeblinks occur. However, as learning progresses, the simple spike frequency decreases substantially during the interval between the conditioned stimulus and the unconditioned stimulus, and thereby drives the expression of a conditioned eyeblink response[72,104,106]. This suppression in simple spike response correlates with the kinematics of the eyelid closure[72,87], is reversible[69] and persists as long as the conditioned responses can be generated[88,135,136]. The suppression appears to be both necessary and sufficient to induce the conditioned response, as optogenetic stimulation or inhibition of the Purkinje cells following application of the conditioned stimulus is sufficient to abolish or evoke the response, respectively[88,134].

The suggestion that simple spike suppression has a dominant role during eyeblink conditioning is also supported by the prominent impact of simultaneously blocking of the inhibition provided by the molecular layer interneurons and the induction of LTD at the parallel fibre to Purkinje cell synapse[72,107]. Both of these mechanisms (that is, the active inhibition and induction of LTD) can partly compensate for each other if necessary, as blocking just one of them in isolation provides a relatively mild or no phenotype[65,72,107,137]. This is in line with the fact that both mechanisms are supported by the increase in climbing fibre activity[56,138] that occurs at the onset of the emerging simple spike suppression[71,72,88,139]. One could consider this increasing climbing fibre activity as a reinforcing,

Fig. 5 | **Learning behaviour controlled by upbound microzones. a** | The vestibulo-ocular reflex (VOR) serves to maintain the stability of images on the retina during head movements. The gain of this reflex (blue trace) can be adapted in response to longer-lasting changes in retinal slip that can occur due to altering feedback of the visual surrounding; for example, by rotation of a drum around a mouse that moves in a direction opposite that of the head, the gain will be increased (upper part of schematic). The lower part of the schematic illustrates the circuitry of the vestibulocerebellum and brainstem through which VOR adaptation occurs (the upbound micromodule is highlighted in blue). Increases in the simple spike activity of Purkinje cells (PCs) in the flocculus induce an increase in the gain of the ipsiversive horizontal VOR by increasing the inhibition onto the inhibitory floccular target neurons (ftn) in the vestibular nuclei and subsequently enhancing the activation of the ipsilateral abducens motor neurons (abd) and lateral rectus muscle; a net excitatory effect is achieved because of the odd number of inhibitory connections downstream of the inhibitory PCs[73]. The increase in simple spike activity is induced by a reduction in activity of climbing fibres originating from the contralateral inferior olive (IO) that signal changes in retinal slip around the vertical axis. Vestibular signals are generated in the semicircular canals: rotation of the head towards the contralateral side drives a decrease in the firing rate of semicircular canal neurons that is relayed via the vestibular ganglion cells and the mossy fibre output from the floccular projection neurons (fpn) to the granule cells (GCs) in the ipsilateral flocculus[69,73]. The vestibular input provides a necessary context for the adaptation to occur, yet the plasticity essential for the initial stages of learning presumably occurs predominantly at the level of the PCs. The vestibular signals are also transferred to the other side of the brainstem via commissural projection neurons (cpn)[119]. The ftn, fpn and cpn together form the group of vestibular nuclei neurons (VNs). **b** | Adaptation of whisker protraction serves to modify whisker reflexes following repetitive external perturbation, such as an increased frequency of air-puff stimulation to the whiskers that activates trigeminal ganglion neurons and subsequently neurons in the trigeminal nucleus of the brainstem[101]. This adaptation is predominantly controlled by upbound microzones embedded in lobule VII of the lateral cerebellar cortex (FIG. 1). Different neurons within the trigeminal nucleus provide a sensory input to the GCs in the cerebellar cortex (via the mossy fibres) and to the neurons in the IO[76,133]. A reduction in climbing fibre activity originating from the subnucleus of the contralateral IO that signals directional changes in whisker position facilitates an increase in simple spike activity of the Purkinje cells in these microzones[101]. These increases in simple spike activity in turn induce an increase in whisker protraction by reducing the activation of the intermediate reticular formation, which inhibits whisker motor neurons residing in the ipsilateral facial nucleus. Thus, here too, the circuit downstream of the inhibitory Purkinje cells entails an odd number of inhibitory connections, although the order in which they appear in the circuit is reversed when compared with that in the vestibulocerebellum. The upbound micromodule controlling whisker protraction is highlighted in blue. CN, cerebellar nucleus neuron. Part **a** adapted from REF.[73], CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/). Part **b** adapted from REF.[101], CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/).

reward-like signal[71], as it is augmented as conditioning progresses[71,72,88,139]. Finally, the effects that occur at the level of the synaptic inputs of the interneuron axons and parallel fibres onto Purkinje cells and suppress their simple spike activity may be enhanced by a metabotropic glutamate receptor 7-dependent mechanism that engages an intrinsic cascade of events[137,140].

The learning-dependent suppression of simple spike activity during eyeblink conditioning aligns with the downstream circuitry (FIG. 6a); the target neurons in the cerebellar nuclei excite premotor neurons in the red nucleus that, in turn, drive the motor neurons in the facial nucleus that are responsible for eyelid closure[141]. This is thus a completely excitatory pathway that will be activated as a whole when Purkinje cell inhibition decreases[88].

Preparation for goal-directed movements based on perception of sensory cues also requires cerebellar processing that needs to be acquired (FIG. 6b). These activities can range from relatively simple forms of motor planning in the medial cerebellum, such as directional tongue movements following object discrimination in the whisker field[78], to more complex forms in more lateral parts of the cerebellum that require, for example, modification of limb or tail movements following visual or sensory discrimination[43,74,79,142] or processing of acquired reward or no-go signals[45,89]. Even though the precise identity of the microzones that regulate these goal-directed behaviours remains to be identified, dominant 'ramping down' changes in simple spike activity in the Purkinje cells and concomitant 'ramping up' changes in neuronal activity in the cerebellar nuclei have been consistently reported[43,78], indicating that at least some involve downbound microzones. This notion is further supported by the fact that the peak in complex spike activity of the Purkinje cells during goal-directed motor planning in the hemisphere perfectly coincides with the trough of their decreasing simple spike activity[43], highlighting the reciprocal firing pattern of their Purkinje cells following effective training[72,91]. Moreover, the downstream effects of these changes in cerebellar activity appear to be mediated via short-latency thalamic pathways to the prefrontal cortex[43,78], which are predominantly excitatory[143] (FIG. 6b).

Microzones operating in synergy

Even though many microzones may exert a dominant, unidirectional impact on a particular behaviour, they probably never operate in isolation. This holds for both upbound and downbound microzones; they can operate in synergy with another microzone of the same identity (FIG. 7a) and/or with one of opposite identity (FIG. 7b). The upbound microzone of Purkinje cells in the flocculus of the vestibulocerebellum that mediates adaptation of the VOR in the ipsilateral direction does not show an increase in simple spike frequency when the gain of the eye motion towards the contralateral side is enhanced[73]; instead, this latter enhancement occurs when the sensitivity of the Purkinje cells on the contralateral side to head rotation towards the ipsilateral side is increased. However, when the contralateral rectus muscle elicits contraversive eye motion, the corresponding muscle on the ipsilateral eye has to relax. This relaxation is supported by commissural connections in the brainstem[144,145] as well as by a decrease in simple spikes on the ipsilateral side during the contraversive VOR[73]. As this latter decrease is less prominent than the increase that occurs during ipsiversive learning[69,146], one can conclude that the two upbound microzones in the left vestibulocerebellum and right vestibulocerebellum operate in synergy, together fine-tuning in a complementary and alternating fashion the enhancement of both ipsiversive and contraversive eye movements (FIG. 7a). Another complementary process occurs when the VOR gain towards the ipsilateral side has to be reduced; this can be induced by installing a permanent visual field that moves with the head[147]. The experimental data suggest that such a gain-decrease adaptation can be mediated largely by plasticity in the vestibular nuclei downstream of the flocculus, rather than by a change in simple spike activity[64,73,110,147–149].

Likewise, the upbound microzones in the lateral cerebellum that control adaptation of whisker protraction

by increasing their simple spike frequency do not show this increase when the animals retract their whiskers[101]. However, here the counterforce required to fine-tune the balance of agonistic and antagonistic functions does not originate from other contralateral upbound microzones (as occurs for adaptation of the VOR) but probably occurs from downbound microzones that are directly adjacent to the upbound microzones[76,101]. Indeed, whereas the upbound microzones involved in whisker protraction show prominent simple spike increments with weak complex spike responses during learning, the downbound microzones appear to show decreases in simple spike activity with strong complex spike responses[101]. The Purkinje cells in the downbound microzones are fewer in number and respond at longer latencies, but appear well designed to control complementary functions such as

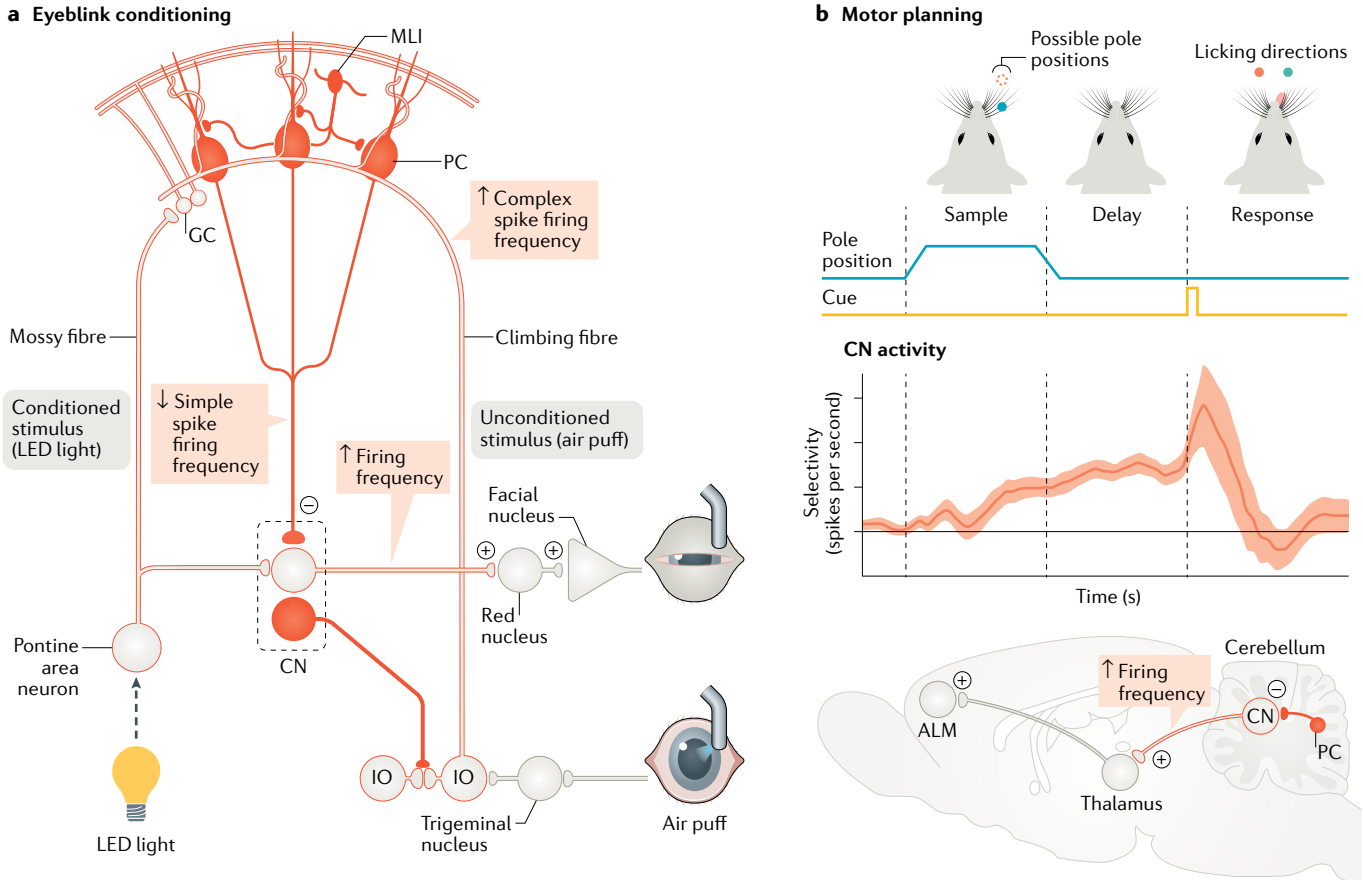


Fig. 6 | **Learning behaviour controlled by downbound microzones.** **a** | Pavlovian eyeblink conditioning permits the generation of new associative sensorimotor memories following the systematic pairing of a neutral conditioned stimulus with an unconditioned stimulus to the eye. In the example illustrated, the conditioned stimulus is provided by an LED light, triggering activity in the pontine area, which relays the corresponding signals via the mossy fibre–granule cell (GC)–parallel fibre pathway to the Purkinje cells (PCs) in the cerebellar cortex. The unconditioned stimulus in this case is provided by an air puff to the eye; this stimulus activates the trigeminal nucleus and inferior olive (IO), via which the climbing fibre signals reach the appropriate downbound microzone of PCs in the deep fissure of lobule VI in the cerebellar cortex[69,72]. Decreases in simple spike frequency facilitated by active inhibition from the molecular layer interneurons (MLIs) drive the expression of a conditioned eyeblink response mediated by excitatory red nucleus and facial nucleus neurons. Thus, as illustrated here, given the net polarity of neurotransmission in their downstream neurons, the simple spike activity of Purkinje cells in downbound microzones must be suppressed to express the conditioned response. Moreover, as training progresses, the onset of complex spike responses shifts towards the onset of the conditioned response, deepening the suppression in the simple spike responses and thereby reinforcing the learning[71,72]. The downbound micromodule controlling eyeblink conditioning is highlighted in red. **b** | The planning for goal-directed movements allows mammals to learn to convert the perception of sensory cues into specific well-timed preferred actions in a particular direction. For example, in the task illustrated in the upper panel, a mouse has to identify the location of a pole in the posterior (blue dot) or anterior (dashed red dot) part of its whisker field during the sample stage and to learn to generate an appropriate tongue movement to the right (blue) or left (red), respectively, during the response stage. During the performance of such a task, cerebellar downbound microzones and microcomplexes (in red) are engaged to facilitate the motor planning[43,78]. Excitatory neurons in the cerebellar nuclei (CNs) ramp up their activity in a correlated manner during the delay stage[78], while they are presumably supported by PCs that concomitantly ramp down their simple spike activity (not shown)[43]. As the mice do not appear to make any related movement during the long-lasting delay stage (more than 1 s)[78], the ramping activity during this period points towards a putatively preparatory cognitive activity. These effects are relayed to the anterolateral part of the cerebral motor cortex (ALM) via excitatory neurons in the thalamus (bottom panel). The downbound microcomplex controlling motor planning is highlighted in red. Thus, here too, the net polarity of synaptic transmissions downstream of the PCs is positive, highlighting the need for downbound changes in simple spike activity to modify behaviour. 'Selectivity' in the middle panel refers to the spike rate difference between the preferred and non-preferred trial type (that is, the difference in firing when the CN is engaged in a trial in which it shows its preferred modulation versus that when it is not engaged in such a trial). Part **a** adapted with permission from REF.[69], Cold Spring Harbour Press. Part **b** adapted from REF.[78], Springer Nature Limited.

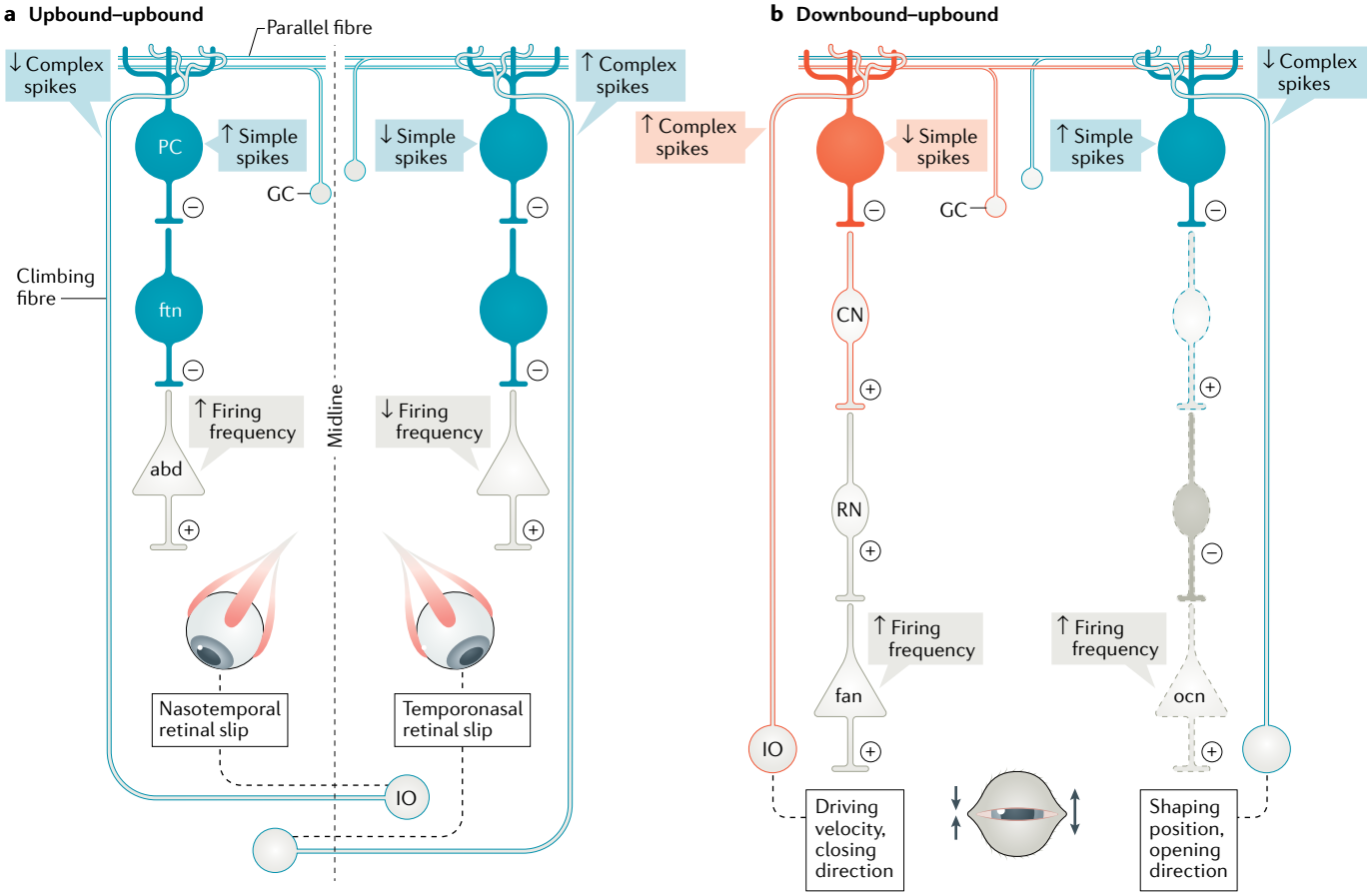


Fig. 7 | **Different microzones can operate in synergy. a** | Upbound microzones in the left and right flocculus of the vestibulocerebellum work together to mediate adaptation of the vestibulo-ocular reflex induced by retinal slip signals relayed via the inferior olive (IO)[73]. When a microzone within one of the flocculi accelerates ipsiversive compensatory eye movements by increasing the simple spike frequency of its Purkinje cells (PCs; see FIG. 5a), the corresponding microzone in the flocculus on the other side shows, albeit less prominently, a decrease in simple spike activity. As a result, the contralateral rectus muscle innervated by the abducens motor neurons (abd) relaxes during an enhanced ipsiversive eye movement and vice versa. Upbound micromodules are indicated in blue. **b** | By contrast, the PCs of the downbound microzone (in red) in lobule VI (or lobulus simplex) that show a prominent simple spike suppression and drive the conditioned eyeblink closure via the cerebellar nuclei neurons (CNs) and facial nucleus neurons (fans) probably work together with PCs in an adjacent smaller upbound microzone (in blue) that show facilitation of their simple spike activity[71,72,87,88] and possibly control oculomotor neurons (ocn) downstream that can open the eye[86]. The latency of the simple spike facilitation of the upbound microzone is longer than that of the simple spike suppression of the downbound microzone[72,88], supporting the closing–opening sequence of events during the expression of the conditioned response. This sequential control by the downbound and upbound modules is in line with the finding that their dynamic changes appear correlated with the speed and position, respectively, of the conditioned eyelid movement[87]. Thus, the two microzones are activated at the same moment by the same conditioned signals mediated by the parallel fibres (FIG. 1b), but due to the different climbing fibre and cerebellar nucleus feedback signals[181,182] as well as the different intrinsic properties of the PCs[58], their simple spike responses are played out sequentially over time and together they facilitate a tailored, synergistic fine-tuning of the differential downstream pathways (part of which remains to be confirmed at the anatomical level, indicated by dashed lines). The examples shown in parts **a** and **b** highlight how simultaneously modulating the parallel fibres innervating different microzones can allow these microzones to work together, fully exploiting the matrix configuration of the architecture of the cerebellum and the micromodular organization of the olivocerebellar system (FIG. 1). RN, red nucleus. Part **a** adapted from REF.[73], CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/).

those responsible for determining the level of contraction in the retraction muscles[133].

The same combination of two microzones of different identity operating in synergy also occurs for behaviours that are dominated by the downbound microzones (FIG. 7b). For example, the downbound microzone in lobule VI containing Purkinje cells that show a prominent simple spike suppression during the expression of a conditioned eyeblink response is bordered by an upbound microzone that shows facilitation of simple spike activity[72,87]. These Purkinje cells are fewer in number and show later and fewer complex spike responses to the unconditioned stimulus than the downbound cells[72,88]. Moreover, the latency of their simple spike facilitation is longer than that of the simple spike suppression of the downbound cells[72], and their profile appears correlated with the position, rather than the speed, of the conditioned eyelid movement[87]. As the activity of these neurons further increases when the eyelid opens up again after the conditioned response[72,88], their main downstream target might be the oculomotor nucleus, which is responsible for raising the eyelid[86]. Thus, here too, a downbound microzone and an upbound microzone may act together, possibly subserving simultaneous

as well as sequential control of antagonistic and complementary functions[86,109,150].

Even though these examples illustrate downbound and upbound microzones simply working together to simultaneously finely regulate complementary and/or antagonistic muscle groups, in principle they may exert such coordinating actions for any complex set of functions, including acceleration and deceleration of locomotion patterns[151,152], regulation of the real-time speed and direction of saccades[36], simultaneous control of autonomic and voluntary functions[131] and even higher cognitive functions such as shifting attention towards specific forms of motor preparation[43,78,153]. Indeed, in the lateral cerebellar cortex one can find downbound and upbound Purkinje cells in neighbouring regions that exhibit high and low levels, respectively, of well-timed complex spike activity during learning-dependent planning[43,46]. It is possible that these different microzones operate in synergy in a sequential, yet coordinated fashion by separately driving the preparation for and execution of movements[45,47,79]. As far as complex spike activity is concerned, one can indeed observe alternating sequences of signals that map onto cerebellar microzones, with preparation and reward delivery often activating different microzones, while suppressing others[46,89] (FIG. 4).

Conclusions and future directions

From the data highlighted herein, the view emerges that the cerebellar cortex is divided into neighbouring upbound and downbound microzones in which the simple spike activity of Purkinje cells increases and decreases, respectively, during learning. In the upbound microzones the initial spontaneous simple spike activity is relatively low, whereas in the downbound microzones this intrinsic activity is high, leaving ample room in both types of zones for changes in the opposite direction during the initial stages of training. The main microzone responsible for controlling a particular behaviour or cognitive process is dominated by the demands of the functional motor network controlling movements on or towards the ipsilateral side of the body, and/or by the requirements of higher processing in the contralateral cerebral cortex. Indeed, it appears possible to predict whether a particular behaviour is primarily controlled by an upbound or a downbound microzone, purely on the basis of the net sum of all excitatory and inhibitory neurotransmissions in the downstream circuitry.

While a particular functional pathway is influenced in the upbound or downbound direction during learning, complementary pathways are frequently manipulated in the opposite direction so as to facilitate graded and smooth control. This can be mediated by analogue microzones on the other side of the cerebellum, by adjacent microzones with opposite actions and/or by diminishing activity in the dominant microzones. By the same token, single microzones can also control several behaviours, provided that the combination of net polarities in synaptic transmission of the downstream circuitries that drive these behaviours allows it. Together, the different sets of microzones may provide complementary mechanisms for entraining any process in the brain that requires coordination at high temporal resolution.

All forms of learning in the upbound and downbound microzones described so far herein can be effectively guided by a decrease and an increase, respectively, in climbing fibre activity derived from the contralateral inferior olive. Consistently, decreases in climbing fibre activity will increase simple spike activity, whereas increases in climbing fibre activity will decrease simple spike activity[37,56,91]. These correlations are congruent with the known mechanisms for synaptic plasticity: that is, all forms of presynaptic and postsynaptic plasticity in the molecular layer of the cerebellar cortex that are synergistically driven by the absence or presence of climbing fibre activity[53]. The precise temporal relationship between the activity in the climbing fibre pathway and that in the mossy fibre–parallel fibre pathway that will optimize the induction of the different forms of plasticity can probably vary from microzone to microzone on the basis of their dedicated intrinsic cell physiological properties[55,56,58,98,154].

The model of bidirectional learning in upbound and downbound microzones highlighted here does, however, raise several questions that will need to be addressed before we will be able to comprehend all its mechanisms and consider it as a general unifying concept of cerebellar function.

Alignment of input stage with output stage. The specific learning properties of the upbound and downbound microzones directly emerge from the circuitry and mechanisms occurring in the Purkinje cell layer and the molecular layer of the cerebellum; that is, the output stage of the cerebellar cortex (FIG. 1). However, it remains to be elucidated to what extent the differential operations of upbound and downbound microzones at the output stage are facilitated by an analogous organization of the input stage; that is, the granular layer. One can find support for spatially aligned arrangements of the inputs to both the upbound and the downbound microzones. For example, the mossy fibres that originate from the pons terminate largely on granule cells that feed with their ascending axons into microzones that are positive for ALDOC[155]. In addition, the parts of the granular layer that project to the upbound microzones of the nodulus and uvula of the vestibulocerebellum contain a variety of classes of unipolar brush cells that can be anatomically linked to the microzonal organization of the molecular layer[156] (FIG. 3a). Likewise, the parts of the granular layer that feed into the downbound microzones of the anterior lobe of the cerebellum comprise a variety of classes of mossy fibre inputs and granule cells that can be anatomically and physiologically linked to specific microzones in the molecular layer[157]. Moreover, the irregularity and bursty character of the firing of mossy fibre inputs is significantly greater in the parts of the granular layer that innervate the mixed upbound and downbound microzones of lobules VI and VII than it is in parts innervating the upbound microzones in lobule X[158,159]. As a difference in the firing pattern of the mossy fibres is propagated through the granule cells to the Purkinje cells and molecular layer interneurons, this may well ultimately affect the Purkinje cell simple spike activity. Thus, even though the differences in the intrinsic

properties of Purkinje cells in the upbound and downbound microzones are very robust[11,97], when deciphering their respective learning rules, one should not forget to take into account differences at their input stage.

Propensity for cellular plasticity. Even though the current bidirectional learning model of upbound and downbound microzones predicts that in the intact cerebellum in vivo the parallel fibre to Purkinje cell synapses within these microzones will be predominantly subject to LTP and LTD, respectively, this does not necessarily imply that the intrinsic physiological properties of these cells show the same propensities. In other words, during in vivo learning the occurrence of particular forms of plasticity in each type of microzone is likely to be dominated by activity in the climbing fibre system and the feedback operations of its micromodular network[139,160], but at the level of single Purkinje cells the capacity for induction of LTP and LTD may be more in balance to avoid ceiling effects[161]. Thus, because Purkinje cells in downbound microzones appear to be more susceptible to the induction of LTD at the parallel fibre to Purkinje cell synapse than those in upbound microzones[154], it is possible that these cells might also contain robust intrinsic mechanisms that permit induction of LTP so as to prevent saturation which will render them unable to absorb any new demand for synaptic change. Importantly, molecular analysis has revealed that Purkinje cells that do or do not express ALDOC (presumably largely reflecting the upbound and downbound microzones, respectively) overexpress differential sets of proteins that determine not only their intrinsic level of excitability (FIG. 3) but also their propensities for plasticity[97]. These include proteins that are probably involved in induction of both LTP and LTD as well as calcium-dependent proteins that play a regulatory role in postsynaptic plasticity[11,57,64,98,154,162–166]. Thus, it will be interesting to find out to what extent the induction of the different forms of plasticity at the various inputs to Purkinje cells diverges between upbound and downbound microzones, both in vivo and in vitro.

Secondary versus initial stages of learning. The learning rules highlighted so far hold for the initial stages of learning, but for some forms of learning the trends in simple spike changes may alter over time. For example, for some forms of VOR adaptation in primates there is an intricate relationship between memory formation in Purkinje cells and the storage of the memory in their target neurons in the vestibular nuclei[110,148]. The Purkinje cells and neurons in the vestibular nuclei are subject to different sets of plasticity mechanisms that play out differently over time, while they can probably also influence one another[59,61,149]. Therefore, during VOR adaptation, changes in the simple spike activity of some of the Purkinje cells in the upbound microzones of the floccular complex appear initially linearly related to the level of behavioural learning[73,118], but eventually the simple spike activity may change its depth of modulation or even reverse its polarity[148]. Whether such reversal mechanisms also exist in downbound microzones remains to be seen. With regard to the examples discussed herein — namely, eyeblink conditioning and motor preparation — the downbound changes in simple spike activity during expression of the learned responses appear to be long lasting[136] and, if anything, tend to become stronger over time, ultimately strengthening the level of rebound potentiation in the cerebellar nuclei[43,72,88,167] (FIG. 4b). Some short-term forms of non-synaptic plasticity, such as intrinsic changes in cellular excitability, that can last for longer durations have been found downstream in the neurons in the vestibular nuclei downstream of upbound microzones[147], while forms of enduring structural plasticity, such as axonal sprouting and reformation of perineuronal nets, can be associated with learning in downbound microzones and microcomplexes[60,66,67]. From a teleological point of view, upbound microzones showing increased simple spike activity only during the initial stages of learning might be advantageous because persistent increased activity could ultimately lead to unnecessary energy demand and/or cytotoxicity[69,168]. In addition, here too, secondary effects in downstream nuclei may compensate for the ceiling effects that might occur if a cell is predominantly subjected to one type of postsynaptic plasticity. As secondary plasticity downstream alleviates the necessity for changes in Purkinje cell modulation, the climbing fibres in the corresponding upbound micromodule will adjust their signals in a graded fashion[42,75,160] and reduce the level of LTP induction through homeostatic network mechanisms[167,169,170]. Future studies will have to further elucidate to what extent and where secondary forms of plasticity occur and to what extent upbound and downbound microcomplexes differ in this respect.

Gain-decrease versus gain-increase learning. Some forms of learning require a stronger change in the activity of the Purkinje cells than others. For example, as highlighted already, gain-increase training of the VOR requires prominent changes in the simple spike activity of Purkinje cells in the upbound microzones of the flocculus, whereas gain-decrease training requires no or little Purkinje cell modulation[64,73,110,148]. Why is this? The most parsimonious explanation is that injecting extra acceleration signals into the vestibular nuclei, as is needed during gain-increase training, requires enhanced simple spike signalling from the Purkinje cells[115]. In other words, to generate extra force at the end of the vestibulo-oculomotor network at the level of the abducens motor neurons innervating the eye muscles, the network requires a cell at its origin (that is, a Purkinje cell in an upbound microzone) that can rapidly alter its activity. In this case, it must increase its activity because of the net polarity of neurotransmissions in the downstream network. Instead, during gain-decrease training, which requires predominantly a reduction rather than an increase in muscle activity, the available machinery for plasticity in the vestibular nuclei of the upbound module appears to be virtually sufficient to accommodate the behavioural requirements, which are less demanding in terms of temporal dynamics[147,149]. Similarly, in the downbound module controlling eyeblink conditioning, quickly closing the eyelid in response to a conditioned stimulus requires a cell at its origin (that is, a Purkinje cell in a downbound microzone) that can

rapidly alter its activity. In this case, it must decrease its activity because of an opposite polarity of neurotransmissions in the downstream network. Even so, unlike VOR gain-decrease adaptation in the upbound microcomplex, actively reducing the amplitude of the eyelid closure in conditioned animals requires a concomitant change in simple spike activity in the relevant downbound microzone in lobule VI (REF.[69]). Indeed, simple spike modulation and eyelid position remain tightly correlated over time during extinction–reacquisition experiments, and the simple spike suppression in the relevant downbound Purkinje cells is apparent as long as conditioned eyeblink responses can be expressed[69]. Possibly, this difference reflects the potential need not only to rapidly close the eyelid but also to rapidly open it afterwards, requiring similar dynamic demands. Thus, it will be interesting to study to what extent upbound and downbound microcomplexes differ in their dynamic control of gain-decrease versus gain-increase paradigms in general.

Sensitivity and firing rate. The bidirectional model suggests that the increases and decreases in simple spike activity that occur during the initial stages of cerebellar learning are predominantly expressed as changes in the firing rate. Such changes often reflect changes in the sensitivity of Purkinje cell synapses to parallel fibre inputs through induction of potentiation or depression[53]. As a result, Purkinje cells can, for example, become more sensitive to a particular peripheral input (such as a proprioceptive, vestibular or visual input) and thereby alter their magnitude sensitivity[110,115,148,171,172]. To what extent postsynaptic plasticity processes in upbound and downbound microzones can change the sensitivity of synaptic inputs to Purkinje cells without changing the simple spike firing rate remains unknown. However, recent studies indicate that this is possible[74,173,174]. For example, when a subject has to track an object while the feedback is constantly manipulated, the sensitivity of simple spike responses to the kinematics of the movements or task performance can be temporarily altered without significant changes in the firing rate[1]. Moreover, these changes in simple spike sensitivity are also bidirectional and guided by changes in climbing fibre activity[1,173]. Hence, it will be interesting to find out to what extent this dynamic control of simple spike sensitivity can meet the demands of upcoming tasks following learning with an altered, yet persistent, feedback, to what extent such putative increases and decreases in simple spike sensitivity are associated with more persistent changes in firing rate and to what extent these bidirectional changes are differentially controlled by upbound and downbound microzones.

Localization of other behaviours. There are many types of motor learning that have been well described in terms of the modulations of simple spike and complex spike activity that occur during behavioural changes but still require investigation with regard to their precise topographic relationship with different microzones. These include mainly types of motor learning studied in non-human primates, including smooth pursuit learning[42,63,175], saccadic adaptation[36,37,176], acquisition of goal-directed limb movements[74,177] and reinforcement learning[79]. In all cases, the predictions of the model articulated herein, including bidirectional control by the presence and the absence of climbing fibres, are upheld; however, histological verification remains to be done to find out which microzones and micromodules are responsible for the coordination of these specific forms of cerebellar learning. Such analyses may also reveal to what extent complementary behavioural encodings from mixed populations of Purkinje cells reflect the synergistic operations of different upbound and downbound microzones[37,178]. It should be noted, however, that the organization of microzones in rodents cannot always be directly extrapolated to that in primates. For example, all microzones of the rodent flocculus appear to express ALDOC[31,69,112], whereas prominent alternating ALDOC-positive and ALDOC-negative microzones are observed in primates[179,180]. In this respect it will also be interesting to find out whether the distribution of ALDOC-negative microzones in primates aligns with downbound simple spike activity during smooth pursuit learning[175], which does not occur in rodents. By the same token, for many of the cerebellar learning behaviours it remains to be established to which downstream circuitries the corresponding micromodules are precisely connected, and whether their main pathways have the net polarity of neurotransmissions, as predicted by the upbound and downbound microzones highlighted above.

Bidirectional control of cognition. It will be interesting to find out which of the cognitive functions that are processed in the cerebral cortex may be controlled by upbound microzones in the cerebellum. As most of the climbing fibre and mossy fibre projections to the lateral cerebellar cortex are derived from the principal olive and pons, respectively[3,13], and as both regions receive prominent monosynaptic or bisynaptic inputs from the cerebral cortex while they project predominantly to ALDOC-positive microzones[69,155], there must be ample room to regulate cerebral cortical functions in upbound microzones. In other words, given that the processing of motor preparation in the anterolateral cortex can be influenced at short latency by downbound activity in both medial and lateral parts of the cerebellar cortex[43,78], one wonders which potential complementary functions may be exerted by the neighbouring upbound microzones. As some of the upbound Purkinje cells in the crus 1 and crus 2 areas of the lateral cerebellum that regulate cerebral cortical activity may be involved in the execution of movements[79,101], it will be interesting to find out whether these upbound and downbound microzones operate in synergy in a sequential fashion by driving the preparation for and execution of movements, respectively[45,47]. One attraction of this hypothesis is that it offers a mechanism to fine-tune the phase and sequence of two stages of a behaviour (that is, preparation and execution) that should not be engaged simultaneously[181,182].

Antagonism and/or complementarity? When different upbound and downbound microzones are activated simultaneously, it is unclear to what extent they exert

Magnitude sensitivity
The ratio of the amplitude of spike modulation to the amplitude of modulation of the position of a moving part of the body.

antagonistic and/or complementary functions. With regard to direct control of motor domains, many motor behaviours already have circuitries in place in the brainstem and/or spinal cord that facilitate simultaneous control of the activation and relaxation of agonistic and antagonistic muscles, respectively, sometimes even subserving alternating rhythmic activity. For example, the internuclear afferents connecting the abducens nucleus with the medial rectus eye muscle warrant relaxation of the medial rectus eye muscle when the abducens nucleus evokes an ipsiversive movement of the eye[144,145,183]. Likewise, the brainstem has extensive circuitries that allow rhythmic, whisking, respiratory or locomotion behaviour, alternatingly driving agonists and antagonistic motor domains[126,128–130,151]. Moreover, even the oculomotor nucleus, which is responsible for opening of the eyelid through activation of the levator palpebrae muscle, has connections via brainstem pathways with the facial nucleus mediating eyelid closure[183]. Why then do we need a superimposed system of cerebellar microzones that controls these brainstem areas? Is it just to fine-tune the agonist and antagonistic functions embedded downstream in the pathway sequentially over time and/or is it actually to simultaneously accelerate the timing of multiple and complementary motor actions? For example, downbound and upbound microzones may leverage the intrinsic capacities of their Purkinje cells to differentially integrate climbing fibre and parallel fibre inputs over time[58] and generate well-timed sequences of motor activities, such as required for closure and opening of the eyelids during eyeblink conditioning[72,88,181,182] (FIG. 7b), or the cerebellar organization of adjacent microzones could help to coordinate head movements with eye movements into certain directions in space[36,37,184], locomotion with facial conditioning[185–187] or whisking with respiration[131]. Direct control through a combined activation of upbound and downbound microzones that simultaneously activate agonistic and relax antagonistic muscle groups would require not only that these microcomplexes are adjacent to one another[86–88] but also that the downstream pathways of these microzones actually have the same net polarity in neurotransmission. Alternatively, they might be engaged simultaneously to control joined tonic contractions of agonistic and antagonistic muscle groups, which can be required for particular motor tasks. Future studies will have to further explore how and when upbound and downbound microzones operate in synergy, and to what extent they serve to control antagonistic and/or complementary functions.

Refinement of microzones. Last but not least, the differential molecular machineries that underlie the cell physiological differences between the upbound and downbound microzones are just starting to be unravelled[11,31,69,93,94,97,98]. As the activity of neurons can only increase or decrease, it is obvious that microzones can be subdivided into two main categories. However, given the individual demands within the upbound or downbound microzones, it will be interesting to correlate the subtleties of the molecular composition of each and every microzone with its specific cellular properties[188], especially in terms of basic membrane potential and excitability, the propensity for intrinsic and postsynaptic plasticity, and the temporal dynamics of the integration of climbing fibre and parallel fibre inputs[58,97,98,189] (FIG. 3c). As 'only' 126 of the 669 genes that are spatially associated with Purkinje cells can be significantly correlated or anticorrelated with ALDOC or PLCB4 expression[97], there is ample room for further subclasses of microzones. Only with such refinement we will be able to fully grasp the rich repertoire that the whole set of microzones of the olivocerebellar system has to offer, not only for coordinating individual behaviours in a bidirectional fashion[69,87] but also for orchestrating a rich symphony of diverse motor, autonomic and cognitive functions[1,131].

Published online 17 November 2020

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Acknowledgements
This work is dedicated to J. I. Simpson, former supervisor of the author. The author is funded by the Dutch Organization for Medical Sciences (ZonMw), Life Sciences (ALW-ENW-Klein), the European Research Council (Advanced and Proof of Concept grants), the EU LISTEN Innovative Training Network programme, the Medical NeuroDelta programme, LSH-NWO (Crossover, INTENSE), Albinism Vriendenfonds NIN, van Raamsdonk fonds, and the Trustfonds of Erasmus University, Rotterdam. He is grateful to J. Kruisbrink and S. Wolff who assisted with the bibliography.

Competing interests
The author declares no competing interests.

Peer review information
Nature Reviews Neuroscience thanks R. Apps, who co-reviewed with J. Pickford; T. Ebner, who co-reviewed with M. Streng; and G. Hesslow for their contribution to the peer review of this work.

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